

# Region-Aware ResNet-18 for Facial Acne Severity Classification

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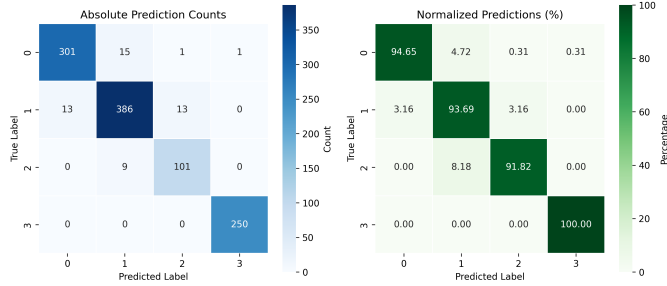


Fig. 1. Test-set confusion matrix (counts and normalized percentages).

## REFERENCES

- [1] K. He *et al.*, “Deep residual learning for image recognition,” CVPR, 2016.

**Abstract**—Acne vulgaris severity assessment remains subjective and specialist-dependent. We present a Region-Aware ResNet-18 that classifies four severity levels from YOLOv8-extracted facial-region crops using learnable region embeddings and a shared backbone. On 1,090 test crops from combined Acne04/Acne Level data, the model reached 95.2% accuracy and weighted F1 of 0.95, with regional F1  $\geq 0.93$ .

**Index Terms**—acne classification, ResNet, computer vision, teledermatology

## I. INTRODUCTION

Automated acne grading can support teledermatology, but multi-stage lesion detectors are costly and error-prone. We propose direct severity classification from region crops with explicit anatomical context encoded by embeddings.

## II. METHOD

Acne04 (1,406 images) and Acne Level (1,457 images) were segmented into five regions with YOLOv8, split 70/15/15, and resized to  $224 \times 224$ . A ResNet-18 backbone (trained from scratch) concatenates 512-D visual features with 64-D region embeddings before a four-class head. Training used Adam, weighted cross-entropy, augmentation, and early stopping.

## III. RESULTS

Global test accuracy was 95.2% (weighted F1 = 0.95). Regional weighted F1: forehead 0.99, chin 0.93, nose 0.97, left cheek 0.93, right cheek 0.96. Figure ?? shows the confusion matrix. Thirty robustness runs averaged 94.41% accuracy ( $\sigma = 0.46\%$ ).

## IV. CONCLUSION

Region-aware embeddings provide anatomical context with a single lightweight CNN. Future work: on-device inference, calibration, and external validation.